Questions

Q1.

Which of the following does **not** use microwaves?

(1)

☐ **A** a mobile phone

☐ **B** WiFi

☐ **C** a speed camera

☐ **D** a TV remote control

(Total for question = 1 mark)

Q2.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which colour of visible light has the longest wavelength?

(1)

☐ **A** blue
☐ **B** green
☐ **C** red
☐ **D** yellow

(Total for question = 1 mark)

Q3.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

A student is investigating the refraction of light.

Figure 5 shows part of the apparatus and the angles to be measured.

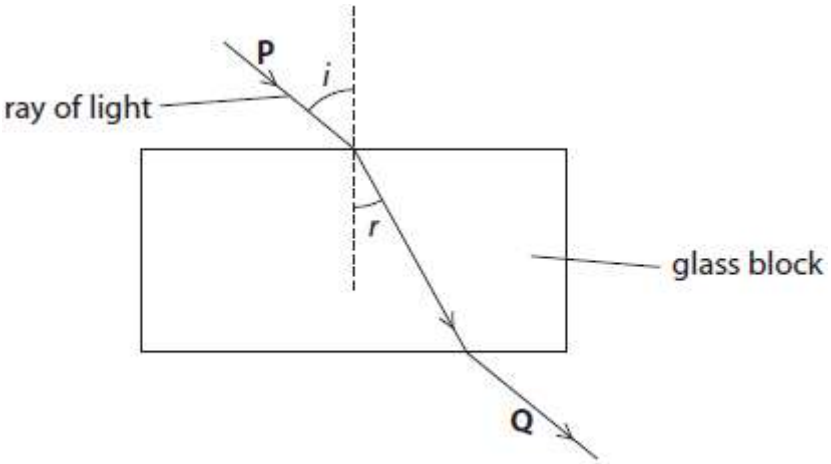
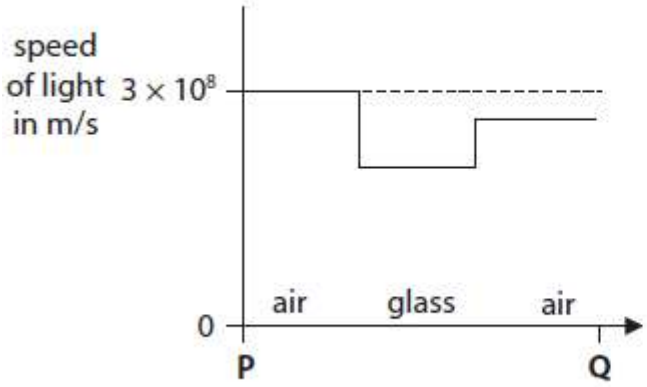


Figure 5

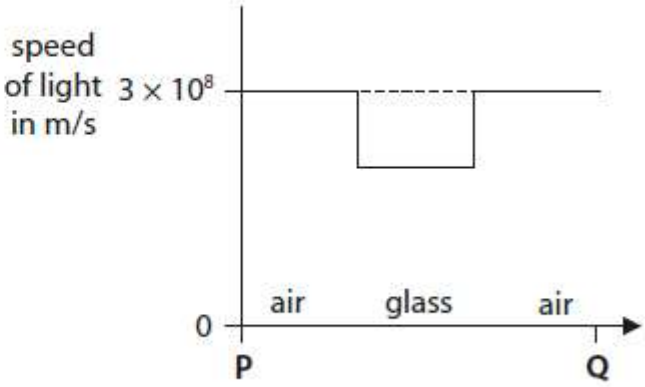
Which of these sketch graphs represents the speed of light as it travels from **P** to **Q**?

(1)

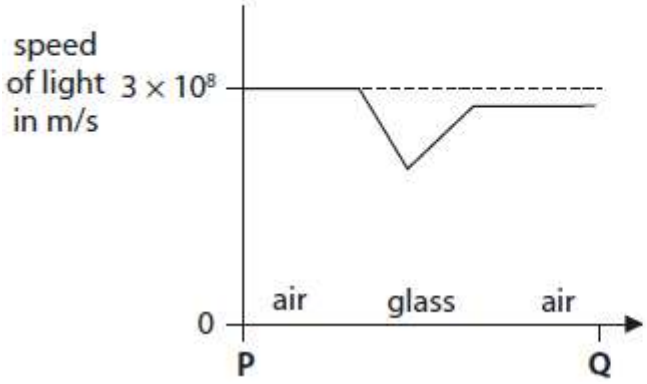
☐ **A**



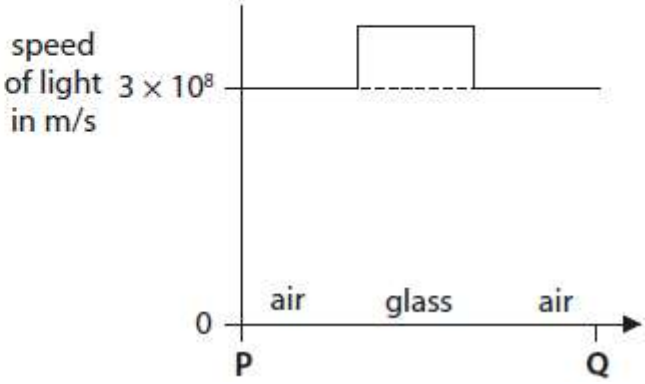
☐ **B**



☐ **C**



☐ **D**



(Total for question = 1 mark)

Q4.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

(i) The law of reflection of light applies to reflections from

(1)

- ☐ **A** all surfaces
- ☐ **B** only shiny surfaces
- ☐ **C** only rough surfaces
- ☐ **D** only smooth surfaces

(ii) A student uses a mirror to demonstrate that the angle of incidence is equal to the angle of reflection.

Figure 8 shows the apparatus the student uses.

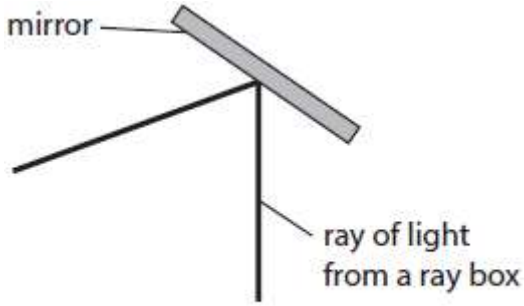


Figure 8

Describe the procedure the student should use with the ray and mirror in the position shown in Figure 8.
You should include any extra equipment needed.
You may add to Figure 8 to help your answer.

(3)

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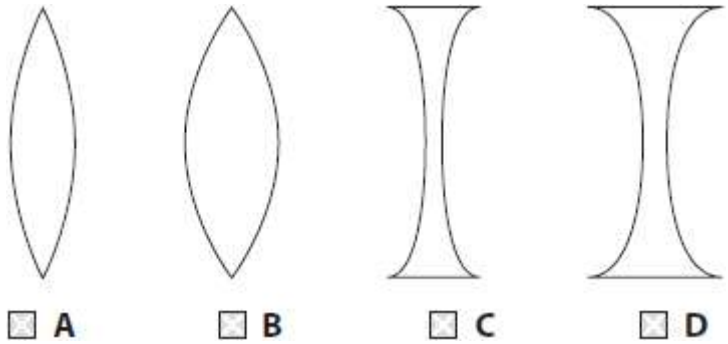
(Total for question = 4 marks)

Q5.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

(i) Which lens is a converging lens with the greatest power?

(1)



(ii) The equation that relates the power of a lens to the focal length of the lens is

$$\text{power (in dioptres)} = \frac{1}{\text{focal length (in metres)}}$$

The power of a lens is 5 dioptres.
Use the equation to calculate the focal length of the lens in cm.

(2)

focal length = ... cm

(Total for question = 3 marks)

Q6.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒

A student does an experiment to determine the critical angle for glass.

The student shines a ray of light into a semicircular glass block and measures the angles *i* and *r*, as shown in Figure 14.

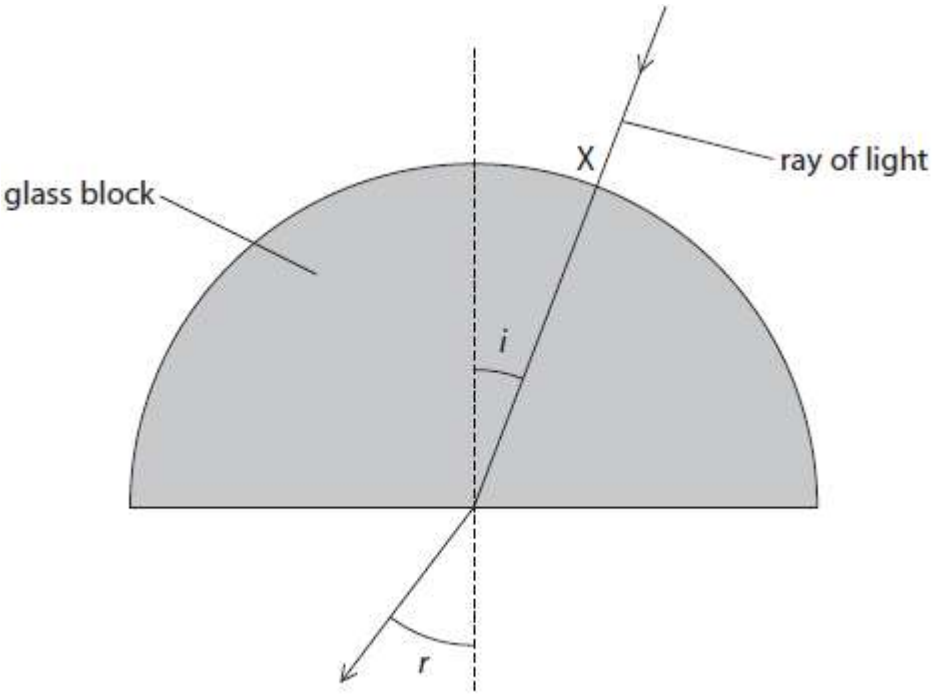


Figure 14

(i) The ray of light does not change direction when it enters the glass block at point **X.**

Which of these **explains** why the ray of light does not change direction when it enters the glass block at point **X.**

(1)

- ☐ **A** The ray enters along a normal to the edge of the block.
- ☐ **B** The ray enters at right angles to a normal to the edge of the block.
- ☐ **C** The ray speeds up as it enters the glass.
- ☐ **D** The ray slows down as it enters the glass.

(ii) The student repeats the procedure for different values of angle *i*.

Figure 15 is a graph of the student's results.

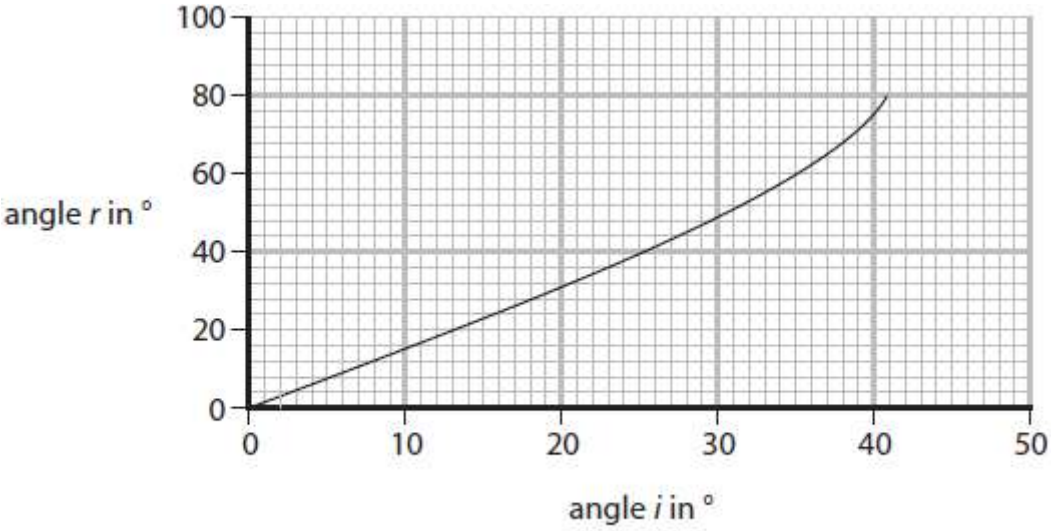


Figure 15

Describe how the student should use the graph in Figure 15 to determine the critical angle for glass.

(3)

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(Total for question = 4 marks)

Q7.

(i) Give **one** example of a virtual image that a lens could produce.

(1)

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(ii) Give **one** example of a real image that a lens could produce.

(1)

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(Total for question = 2 marks)

Q8.

The Asteroid Belt is part of our Solar System.

Vesta is an asteroid in the Asteroid Belt.

Energy is transferred from the Sun to Vesta by radiation.

Explain why the temperature on Vesta does not continue to rise, even though it is absorbing energy from the Sun.

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(Total for question = 3 marks)

Q9.

The asteroid Ceres orbits the Sun between Mars and Jupiter.

During its orbit, Ceres receives electromagnetic radiation from the Sun.

The average temperature of Ceres is constant.

Explain why the average temperature of Ceres remains constant.

(4)

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(Total for question = 4 marks)

Q10.

Figure 16 shows two iron spheres, **P** and **Q**, near to a radiant heater.

P is painted black and **Q** is painted white.

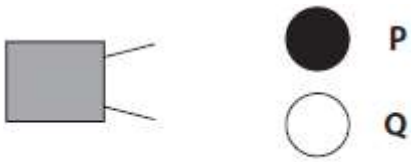

Figure 16

Each sphere is the same distance away from the heater.

The spheres have the same radius.

The heater is switched on and the spheres heat up.

The temperature of each sphere is monitored.

(i) Explain why the temperature of sphere **P** increases at a faster rate than the temperature of sphere **Q**.

(2)

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(ii) The heater remains switched on.

Figure 17 shows how the temperature of sphere **P** changes with time.

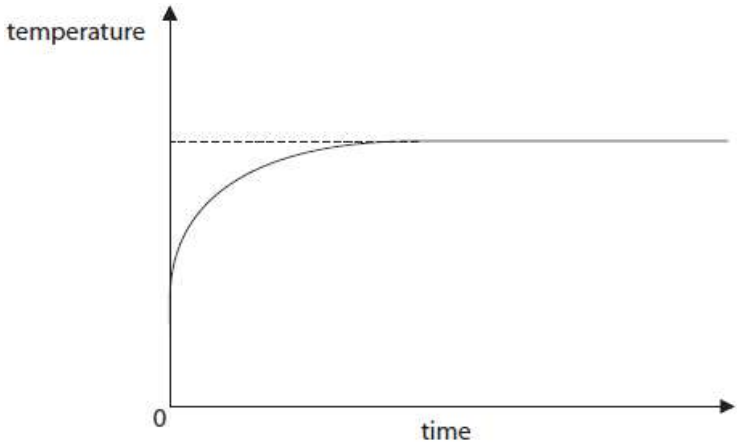

Figure 17

Explain why the temperature of **P** reaches a constant value, even though the heater remains switched on.

(4)

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(Total for question = 6 marks)

Mark Scheme

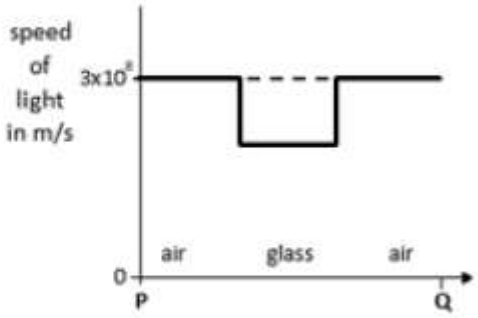
Q1.

Question number	Answer	Additional guidance	Mark
	D a TV remote control		**(1)**

Q2.

Question Number	Answer	Mark
	C red **The only correct answer is C red** **A** is not correct because blue has a shorter wavelength than red **B** is not correct because green has a shorter wavelength than red **D** is not correct because yellow has a shorter wavelength than red	**(1)**

Q3.

Question number	Answer	Mark
CS3	 [X] B **B** is the correct answer	(1) AO3
	A is not correct because the velocity in air is the same before and after the glass C is not correct because the velocity is constant inside the glass D is no correct because the velocity in glass is not greater than the velocity in air	

Q4.

Question	Answer	Mark
(i)	**A** all surfaces the laws of reflection always apply.	1 AO1.2

Question	Answer	Additional guidance	Mark
(ii)	an explanation linking **three** from mark the position of the mirror / mark the position of the rays of light/ place all apparatus on a sheet of paper (1) put in the normal to the mirror which is perpendicular to mirror (at the point of reflection) (1) identify/label angle of incidence and angle of reflection from the normal (1) use a protractor (to measure the angles of incidence / reflection / to find the normal) (1)	marks can be given for additions to diagram may be shown on diagram may be shown as *i* and *r* on diagram do not allow a mark for the use of angle of refraction	**3** **AO3.3**

Q5.

Question Number	Answer	Mark
(i)	 ☐ B **The only correct answer is B** **A** is not correct because it has a smaller power than B **C** is not correct because it is a diverging lens **D** is not correct because it is a diverging lens	**(1)**

Question Number	Answer	Additional guidance	Mark
(ii)	rearrangement and substitution (1) $\frac{1}{5}$ unit conversion and evaluation (1) 20 (cm)	 award full marks for the correct answer without working accept 0.2 for one mark only	**(2)**

Q6.

	Answer	Mark
(i)	**A** The ray enters along a normal to the edge of the block. **B, C** and **D** are incorrect as these do not explain why the light ray does not change direction.	(1) AO2

	Answer	Additional guidance	Mark
(ii)	explanation linking: extrapolate / extend the graph/curve (1) (until it reaches) $r = 90^{o}$ (1) read corresponding value of i (1)	Allow annotation on graph: extension of line on graph to at least $r = 90^{o}$ line down from line to x axis and labelled as critical angle	(3) AO2

Q7.

Question number	Answer	Additional guidance	Mark
(i)	**one** from (with converging lens when used as) a magnifying glass (1) image of an object in a diverging lens (1)	 (when) object is closer to lens than focal length do not accept telescope / microscope / glasses	(1) AO2.1

Question number	Answer	Additional guidance	Mark
(ii)	**one** from image of a distant object (by a converging lens) (1) photograph / image in a camera (1) projector image (1) image in eye / on retina (1)	 (when) object is further from lens than focal length ignore phone	**(1)** **AO2.1**

Q8.

	Answer	Additional guidance	Mark
	An explanation linking: Vesta (also) radiates (energy) (1) the same amount of energy that it absorbs (1) in the same time (1)	 allow emits (OWTE)for radiates 'Vesta radiates at the same average power that it absorbs' scores all three MPs 'Vesta radiates energy at the same rate that it absorbs' scores all three MPs	**(3)** **AO1**

Q9.

Question number	Answer	Additional guidance	Mark
	explanation linking **four** from:		**(4) AO2.1**
	Ceres is absorbing radiation/(thermal) energy from the Sun (1)	allow heat in this context energy is transferred to ceres ceres gains energy (the Sun is) heating it ignore receiving (in stem)	
	Ceres is (also) emitting radiation/(thermal) energy (1)	allow giving out / releasing / losing / dissipating for emitting	
	(rate of) emission increases with temperature (1)		
	(temperature constant) when (rate of) emission = (rate of) absorption (1)		
	mention of rate in either mp3 or mp4 (1)	accept power for rate temperature of Ceres constant when rate of emission of radiation = (rate of) absorption of radiation scores 4 marks absorption of radiation is greater than emission of radiation (so temperature increases) scores MP1 and 2	

Q10.

	Answer	Additional guidance	Mark
(i)	explanation linking any 2 of: black / P is good absorber / poor reflector (1) white / Q is good reflector poor absorber (1) different colour of surfaces absorb differently (1)	allow any one statement for 2 marks: • black / P **better** absorber • black / P **worse** reflector • white / Q **worse** absorber • white / Q **better** reflector • black / P absorbs (radiation) **and** white / Q reflects (radiation)	**(2)** **AO1**

	Answer	Additional guidance	Mark
(ii)	explanation linking **four** from: P is absorbing radiation/(thermal) energy (1) P is (also) emitting radiation/(thermal) energy (1) (rate of) emission increases with temperature (1) (temperature constant) when (rate of) emission = (rate of) absorption (1) mention of rate in either mp3 or mp4 (1)	allow heat in this context temperature of P constant when rate of emission of radiation = (rate of) absorption of radiation scores 4 marks (rate of) absorption of radiation is greater than (rate of) emission of radiation (so temperature increases) scores MP1 and 2	**(4)** **AO1**